ScienceDirect®

Find articles with these terms

review AND (wildfire OR "wild fire" OR "wildland fire") AND (behavior OR dynamics OR tr

Advanced search

FEEDBACK

Loading results ...

relevance | date

Research article

Simplified coupled atmosphere-fire model for simulation of 2D wildland fires

Computer Physics Communications, October 2025

Daniel San Martin, Claudio E. Torres

Research article

Fuel modelling in terrestrial ecosystems: An overview in the context of the development of an object-orientated database for wild fire analysis

Ecological Modelling, 10 November 2009

V. Krivtsov, O. Vigy, ... G. B. Pezzatti

Want a richer search experience?

Sign in for article previews, additional search fields & filters, and multiple article download & export options.

Sign in

Research article *Open access*

FastFuels: Advancing wildland fire modeling with high-resolution 3D fuel data and data assimilation

Environmental Modelling & Software, January 2025

Anthony Marcozzi, Lucas Wells, ... J. Kevin Hiers

View PDF

Research article *Open access*

Computational analysis of wildland fire resistance for transmission line tower

Journal of Constructional Steel Research, January 2025

Chang Chen, Hamid R. Valipour, ... Donghong Zhang

View PDF

Research article *Open access*

Computational assessment of the impact that wildland fire smoke exposure has on airflow and particle deposition in the mouse respiratory tract

Computers in Biology and Medicine, September 2025

Matthew J. Eden, Jacqueline Matz, ... Jessica M. Oakes

View PDF

Research article

Simulations of flaming combustion and flaming-to-smoldering transition in wildland

fire spread at flame scale

Combustion and Flame, April 2024

Mohamed Mohsen Ahmed, Arnaud Trouvé, ... Mark Finney

Research article *Open access*

On using dynamical seasonal forecasts to develop management-driven wildland fire outlooks in Alaska

Climate Services, August 2025

Cecilia Borries-Strigle, Uma S. Bhatt, ... Robert H. Ziel

View PDF

Research article *Open access*

Performance of Vehicle Add-on Mobile Monitoring System PM2.5 measurements during wildland fire episodes

Environmental Science: Atmospheres, 14 March 2024

Ashley S. Bittner, Amara L. Holder, ... William Mitchell

Research article *Open access*

Wildland fire mid-story: A generative modeling approach for representative fuels

Environmental Modelling & Software, January 2024

Grant Hutchings, James Gattiker, ... Rodman R. Linn

View PDF

Research article *Open access*

Kernel density change: A new bitemporal lidar metric for directly mapping wildland fire fuel consumption

Science of Remote Sensing, December 2025

Michael J. Campbell, Andrew T. Hudak, ... Philip E. Dennison

View PDF

Research article

Wildland fire fuels database for Corsican - Mediterranean Forest stand types

Forest Ecology and Management, 1 August 2024

Y. Pérez-Ramirez, L. Ferrat, J. B. Filippi

Research article

Multi-scale 3D cellular automata modeling: Application to wildland fire spread

Chaos, Solitons & Fractals, November 2022

M. Byari, A. Bernoussi, ... M. Amharref

Research article

Multiple social and environmental factors affect wildland fire response of full or less-than-full suppression

Journal of Environmental Management, February 2024

Molly C. Daniels, Kristin H. Braziunas, ... Adena R. Rissman

Research article *Open access*

Advancing the community health vulnerability index for wildland fire smoke exposure

Science of The Total Environment, 1 January 2024

Jihoon Jung, Joseph L. Wilkins, ... June T. Spector

View PDF

Research article *Open access*

Using climate-fire analog mapping to inform climate change adaptation strategies for wildland fire in protected areas of the conterminous US

Global Environmental Change Advances, July 2024

Adam Terando, Peng Gao, ... J. Kevin Hiers

View PDF

Research article *Open access*

A Web-Based Total Worker Health Intervention for Those Fighting Wildland Fires: Mixed Methods Development and Effectiveness Trial

Journal of Medical Internet Research, 2023

Kerry Kuehl, Diane Elliot, ... Bharti Garg

Research article

A novel method for estimation of wild fire intensity based on ash pH and soil microarthropod community

Pedobiologia, 2001

Nava Henig-Sever, Dina Poliakov, Meir Broza

Research article *Open access*

Using wildland fire smoke modeling data in gerontological health research (California, 2007–2018)

Science of The Total Environment, 10 September 2022

Patricia D. Koman, Michael Billmire, ... Sue Anne Bell

View PDF

Research article *Open access*

Rethinking the focus on forest fires in federal wildland fire management: Landscape patterns and trends of non-forest and forest burned area

Journal of Environmental Management, 1 February 2023

Michele R. Crist

View PDF

Research article

Mathematical modeling of wildland fire initiation and spread

Environmental Modelling & Software, March 2020

Vladimir Agranat, Valeriy Perminov

Research article *Open access*

Improving wildland fire spread prediction using deep U-Nets

Science of Remote Sensing, December 2023

Fadoua Khennou, Moulay A. Akhloufi

View PDF

Review article

Mathematical models and calculation systems for the study of wildland fire behaviour

Progress in Energy and Combustion Science, 2003

E Pastor, L Zárate, ... J Arnaldos

Short communication

Are the applications of wildland fire behaviour models getting ahead of their evaluation again?

Environmental Modelling & Software, March 2013

Martin E. Alexander, Miguel G. Cruz

Research article *Open access*

Minimize the bad days: Wildland fire response and suppression success

Rangelands, June 2022

Katherine Wollstein, Casey O'Connor, ... Rod Hoagland

View PDF

Research article

The wildland fire system and challenges for engineering

Fire Safety Journal, March 2021

Mark A. Finney

Research article

Computational modeling of extreme wildland fire events: A synthesis of scientific understanding with applications to forecasting, land management, and firefighter safety

Journal of Computational Science, October 2020

Janice L. Coen, W. Schroeder, ... L Tarnay

Research article

Emulation of wildland fire spread simulation using deep learning

Neural Networks, September 2021

Frédéric Allaire, Vivien Mallet, Jean-Baptiste Filippi

Research article

Computational modeling of extreme wildland fire events: A synthesis of scientific understanding with applications to forecasting, land management, and firefighter safety

Journal of Computational Science, September 2020

Janice L. Coen, W. Schroeder, ... L. Tarnay

Research article *Open access*

One-class Classification-Based Machine Learning Model for Estimating the Probability of Wildfire Risk

Procedia Computer Science, 2023

Fathima Nuzla Ismail, Shanika Amarasoma

View PDF

Short communication

Assessing improvements in models used to operationally predict wildland fire rate of spread

Environmental Modelling & Software, July 2018

Miguel G. Cruz, Martin E. Alexander, ... Musa Kilinc

Research article

An experimental investigation into the wildland fire burning characteristics of loblolly pine needles

Fire Safety Journal, December 2021

Daniel C. Stubbs, Luke H. Humphreys, ... David E. Scarborough

Review article

Fifty years of wildland fire science in Canada

Fifty years of wildland fire science in Canada

Canadian Journal of Forest Research, 18 December 2020

Sean C. P. Coogan, Lori D. Daniels, ... B. Mike Wotton

Research article

Combined estimation of fire perimeters and fuel adjustment factors in FARSITE for forecasting wildland fire propagation

Fire Safety Journal, September 2020

Tengjiao Zhou, Long Ding, ... Zheng Wang

Review article

Uncertainty and risk in wildland fire management: A review

Journal of Environmental Management, August 2011

Matthew P. Thompson, Dave E. Calkin

Research article

Wildland fire emissions, carbon, and climate: Plume rise, atmospheric transport, and chemistry processes

Forest Ecology and Management, 1 April 2014

Warren E. Heilman, Yongqiang Liu, ... Robert Mickler

Research article

Wildland fire smoke and human health

Science of The Total Environment, 15 May 2018

Wayne E. Cascio

Research article

The health impacts and economic value of wildland fire episodes in the U.S.: 2008–2012

Science of The Total Environment, 1 January 2018

Neal Fann, Breanna Alman, ... Ana G. Rappold

Research article

Influence of uncertainties in burned area estimates on modeled wildland fire PM2.5 and ozone pollution in the contiguous U.S.

Atmospheric Environment, October 2018

Shannon N. Koplitz, Christopher G. Nolte, ... James Beidler

Research article

Simulation study of radionuclide atmospheric transport after wildland fires in the

Chernobyl Exclusion Zone in April 2020

Atmospheric Pollution Research, March 2021

Mykola Talerko, Ivan Kovalets, ... Olexandr Romanenko

Review article *Open access*

Peatland-fire interactions: A review of wildland fire feedbacks and interactions in Canadian boreal peatlands

Science of The Total Environment, 15 May 2021

K. Nelson, D. Thompson, ... L. Chasmer

View PDF

Research article

Wildland fire emissions, carbon, and climate: Science overview and knowledge needs

Forest Ecology and Management, 1 April 2014

William T. Sommers, Rachel A. Loehman, Colin C. Hardy

Research article *Open archive*

Novel method for a posteriori uncertainty quantification in wildland fire spread simulation

Applied Mathematical Modelling, February 2021

Frédéric Allaire, Vivien Mallet, Jean-Baptiste Filippi

View PDF

Research article

Factors that contribute to trustworthiness across levels of authority in wildland fire incident management teams

International Journal of Disaster Risk Reduction, 15 April 2022

Claire E. Rapp, Robyn S. Wilson

Research article

Changes in potential wildland fire suppression costs due to restoration treatments in Northern Arizona Ponderosa pine forests

Forest Policy and Economics, February 2018

Ryan A. Fitch, Yeon Su Kim, ... Joe E. Crouse

Research article *Open access*

PDE solution to UAV/UGV trajectory planning problem by spatio-temporal estimation during wildfires

Chinese Journal of Aeronautics, May 2021

Mohammadreza RADMANESH, Balaji SHARMA, ... Donald FRENCH

View PDF

Research article

Progress towards and barriers to implementation of a risk framework for US federal wildland fire policy and decision making

Forest Policy and Economics, June 2011

David C. Calkin, Mark A. Finney, ... Krista M. Gebert

Research article

State-parameter estimation approach for data-driven wildland fire spread modeling: Application to the 2012 RxCADRE S5 field-scale experiment

Fire Safety Journal, April 2019

Cong Zhang, Annabelle Collin, ... Mélanie C. Rochoux

Review article

The myth of the biological threshold: A review of biological responses to soil heating associated with wildland fire

Forest Ecology and Management, 15 January 2019

Melissa R. A. Pingree, Leda N. Kobziar

Review article

Wildland fire risk research in Canada

Environmental Reviews, 26 November 2019

Lynn M. Johnston, Xianli Wang, ... Mike D. Flannigan

Research article

A wildland fire model with data assimilation

Mathematics and Computers in Simulation, 1 December 2008

Jan Mandel, Lynn S. Bennethum, ... Anthony Vodacek

Research article

An assessment of the effects of forest-related policies upon wildland fires in the European Union: Applying the subsidiarity principle

Forest Policy and Economics, April 2013

Luis Galiana, Susana Aguilar, Andrea Lázaro

Research article

Physical model of wildland fire spread: Parametric uncertainty analysis

Combustion and Flame, July 2020

Xieshang Yuan, Naian Liu, ... Domingos X. Viegas

Research article

Wildland fire emissions, carbon, and climate: Wildfire–climate interactions

Forest Ecology and Management, 1 April 2014

Yongqiang Liu, Scott Goodrick, Warren Heilman

Research article

A physical model for wildland fires

Combustion and Flame, December 2009

J. H. Balbi, F. Morandini, ... F. Rinieri

Research article

Sensitivity analysis and parameter adjustment in a simplified physical wildland fire model

Advances in Engineering Software, December 2015

D. Prieto, M. I. Asensio, ... J. M. Cascón

Research article

Wildland fire risk and social vulnerability in the Southeastern United States: An exploratory spatial data analysis approach

Forest Policy and Economics, January 2011

Cassandra Johnson Gaither, Neelam C. Poudyal, ... Jianbang Gan

Research article *Open archive*

Modeling wildland fire propagation with level set methods

Computers & Mathematics with Applications, April 2009

V. Mallet, D. E. Keyes, F. E. Fendell

View PDF

Research article

Physics-informed neural networks for parameter learning of wildfire spreading

Computer Methods in Applied Mechanics and Engineering, 1 February 2025

K. Vogiatzoglou, C. Papadimitriou, ... K. Ampountolas

Research article

Monte Carlo estimation of thermal radiation from wildland fires

International Journal of Thermal Sciences, January 2011

Y. Billaud, A. Kaiss, ... B. Porterie

Correspondence

Commentary on the article “Burn probability simulation and subsequent wildland fire activity in Alberta, Canada – Implications for risk assessment and strategic planning” by J.L. Beverly and N. McLoughlin

Forest Ecology and Management, 15 March 2020

Marc-André Parisien, Alan A. Ager, ... Ellen Whitman

Research article

The influence of risk attitudes on suppression spending and on wildland fire program budgeting

Forest Policy and Economics, April 2020

David Rossi, Olli-Pekka Kuusela

Research article *Open access*

Modeling the interaction between wildfires and windthrows: A pilot case study for Italian Alps

Journal of Environmental Management, December 2024

Luca Mauri, Flavio Taccaliti, Emanuele Lingua

View PDF

Research article *Open access*

Assimilation of Perimeter Data and Coupling with Fuel Moisture in a Wildland Fire–Atmosphere DDDAS

Procedia Computer Science, 2012

Jan Mandel, Jonathan D. Beezley, ... Minjeong Kim

View PDF

Research article *Open access*

Thermally-induced release of arsenic from minerals and phases relevant to polluted natural systems affected by wildfires

Applied Geochemistry, March 2025

Marek Tuhý, Vojtěch Ettler, ... Andrew L. Sullivan

View PDF

Research article

The impact of wildland fires on calcareous Mediterranean pedosystems (Sardinia, Italy) –

An integrated multiple approach

Science of The Total Environment, 15 May 2018

Gian Franco Capra, Simona Tidu, ... Peter Filzmoser

Research article *Open access*

Simulating the impacts of regional wildfire smoke on ozone using a coupled fire-atmosphere-chemistry model

Atmospheric Environment, 1 November 2025

Derek V. Mallia, Cambria White, ... Adam K. Kochanski

View PDF

Research article

Influence of Wildland Fire Along a Successional Gradient in Sagebrush Steppe and Western Juniper Woodlands

Rangeland Ecology & Management, November 2013

Eva K. Strand, Stephen C. Bunting, Robert F. Keefe

Research article

Wildland fire emissions, carbon, and climate: Wildland fire detection and burned area in the United States

Forest Ecology and Management, 1 April 2014

Wei Min Hao, Narasimhan K. Larkin

Review article *Open access*

Forest fire management, funding dynamics, and research in the burning frontier: A comprehensive review

Trees, Forests and People, June 2024

Himanshu Bargali, Aseesh Pandey, ... V. P. Uniyal

View PDF

Research article

The development and implementation of a human-caused wildland fire occurrence prediction system for the province of Ontario, Canada

Canadian Journal of Forest Research, 18 December 2020

Douglas G. Woolford, David L. Martell, ... Dennis Boychuk

Research article

Iterated local search for the placement of wildland fire suppression resources

European Journal of Operational Research, 1 February 2023

André Bergsten Mendes, Filipe Pereira e Alvelos

Research article

Wildland fire emissions, carbon and climate: Characterizing wildland fuels

Forest Ecology and Management, 1 April 2014

David R. Weise, Clinton S. Wright

Research article

Global wildland fire season severity in the 21st century

Forest Ecology and Management, 15 April 2013

Mike Flannigan, Alan S. Cantin, ... Lynn M. Gowman

Review article

Wildland fire ash: Production, composition and eco-hydro-geomorphic effects

Earth-Science Reviews, March 2014

Merche B. Bodí, Deborah A. Martin, ... Jorge Mataix-Solera

Research article *Open access*

Modernizing the US National Fire Danger Rating System (version 4): Simplified fuel models and improved live and dead fuel moisture calculations

Environmental Modelling & Software, October 2024

W. Matt Jolly, Patrick H. Freeborn, ... Stuart Brittain

View PDF

Review article *Open access*

Navigating the evolving landscape of wildfire management: A systematic review of decision support tools

Trees, Forests and People, June 2024

Tristan O'Mara, Andrew Sánchez Meador, ... Elvy Barton

View PDF

Review article

Chapter 6 Airborne Remote Sensing of Wildland Fires

Developments in Environmental Science, 2008

Philip J. Riggan, Robert G. Tissell

Review article

Chapter 4 Chemical Composition of Wildland Fire Emissions

Developments in Environmental Science, 2008

Shawn P. Urbanski, Wei Min Hao, Stephen Baker

Research article

Burn probability simulation and subsequent wildland fire activity in Alberta, Canada – Implications for risk assessment and strategic planning

Forest Ecology and Management, 1 November 2019

Jennifer L. Beverly, Neal McLoughlin

Research article *Open access*

Potential impact of the 2023 Lahaina wildfire on the marine environment: Modeling the transport of ash-laden benzo[a]pyrene and pentachlorophenol

Science of The Total Environment, 1 December 2024

C. A. Downs, Shadan Nasseri Doust, ... Michael J. Risk

View PDF

Review article

Chapter 3 Characterizing Sources of Emissions from Wildland Fires

Developments in Environmental Science, 2008

Roger D. Ottmar, Ana Isabel Miranda, David V. Sandberg

Research article

Modeling topographic influences on fuel moisture and fire danger in complex terrain to improve wildland fire management decision support

Forest Ecology and Management, 15 December 2011

Zachary A. Holden, W. Matt Jolly

Research article Full text access

Operational Application of Mathematical Modeling for Enhanced Wildfire Risk Management in Liguria

IFAC-PapersOnLine, 2024

Francesco Baghino, Emanuele Gissi, ... Paolo Fiorucci

View PDF

Research article *Open access*

Comparative analysis of wildfire simulation tools: Discrepancies in Rothermel model-based software under varying wind and slope conditions

International Journal of Disaster Risk Reduction, 15 April 2025

Shahab Mohammad Beyki, António Gameiro Lopes, ... Aldina Santiago

View PDF

Research article

An interpretable wildfire spreading model for real-time predictions

Journal of Computational Science, December 2024

K. Vogiatzoglou, C. Papadimitriou, ... V. Bontozoglou

Book chapter

Chapter 11: Wildland Fire Danger Rating and Early Warning Systems

Wildfire Hazards, Risks and Disasters, 2015

William J. de Groot, B. Michael Wotton, Michael D. Flannigan

Research article

A comparison of landscape fuel treatment strategies to mitigate wildland fire risk in the urban interface and preserve old forest structure

Forest Ecology and Management, 31 March 2010

Alan A. Ager, Nicole M. Vaillant, Mark A. Finney

Research article

Information systems in support of wildland fire management decision making in Canada

Computers and Electronics in Agriculture, December 2002

B. S Lee, M. E Alexander, ... P Englefield

Research article

Thermal decomposition of vegetative fuels and the impact of measured variations on simulations of wildfire spread

Fire Safety Journal, May 2023

Isaac T. Leventon, Jiuling Yang, Morgan C. Bruns

Research article *Open access*

Assessing telecommunication vulnerabilities to wildfire in Alberta, Canada

International Journal of Disaster Risk Reduction, 15 October 2025

Carter S. Kuiper, Jennifer L. Beverly

View PDF

Research article *Open access*

Features of turbulence during wildland fires in forested and grassland environments

Agricultural and Forest Meteorology, 15 July 2023

Ajinkya Desai, Warren E. Heilman, ... Tirtha Banerjee

View PDF

Conference abstract

03/02639 Mathematical models and calculation systems for the study of wildland fire behaviour: Pastor, E. et al. Progress in Energy and Combustion Science, 2003, 29, (2), 139–153

Fuel and Energy Abstracts, November 2003

No authors available

Research article *Open access*

The response of the invasive princess tree (Paulownia tomentosa) to wildland fire and other disturbances in an Appalachian hardwood forest

Global Ecology and Conservation, September 2021

Angela R. Chongpinitchai, Roger A. Williams

View PDF

Research article *Open access*

Improving wildfire simulation accuracy using satellite active fire data for interval reinitialization and rate of spread adjustment

Remote Sensing Applications: Society and Environment, August 2025

Shahab M. Beyki, António Manuel Gameiro Lopes, ... Luís Laím

View PDF

Research article *Open access*

Human behaviour in fire: Knowledge foundation and temporal evolution

Fire Safety Journal, March 2024

Milad Haghani, Ruggiero Lovreglio, ... Erica Kuligowski

View PDF

Review article

Review of wildfire modeling considering effects on land surfaces

Earth-Science Reviews, October 2023

Dani Or, Eden Furtak-Cole, ... Sean A. McKenna

Research article *Open access*

Experimental and numerical investigation of burning characteristics of Australian pine trees for wildfire modelling

Fire Safety Journal, November 2025

Siddharth Gupta, Mohamed Sharaf, ... Khalid Moinuddin

View PDF

Review article

A review of post-wildfire adaptations of surface-water-quality models: Synthesis, gaps, and opportunities

Science of The Total Environment, 1 June 2025

Zachary M. Shephard, Trevor F. Partridge, ... Brian A. Ebel

Conference abstract

Wildland fire risk—Integrating community resilience or community vulnerability attributes and hazard assessments, to provide a comprehensive risk model

Forest Ecology and Management, 15 November 2006

Klaus Braun

Research article

Foliar moisture content of Pacific Northwest vegetation and its relation to wildland fire behavior

Forest Ecology and Management, 15 August 2002

James K. Agee, Clinton S. Wright, ... Mark H. Huff

Display: 25 | 50 | 100 results per page

previous | Page 2 of 10 | next

ELSEVIER

All content on this site: Copyright © 2025 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

RELX™